CA-3 CASE STUDY

Big Data Analytics for LHC Event Quality and Exploratory Outlier Ranking in 40 MHz Training Data

Big Data Analytics CA-3 Case Study Presentation

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Hadoop Hive Spark 3.5.7 Pig 0.17 scikit-learn 1.6.1

Agenda

Phase 1 – Foundation & Ingestion

1. **Introduction** (Problem & Deliverables)
2. **Dataset & Schema** (Zenodo & 13.45M Rows)
3. **Pipeline** (Architecture & Scaling)
4. **Quality & Distributions** (Zero-Defect Audit)

Phase 2 – Computation & Insights

5. **Spark Analytics** (9 SQL Queries & RDD)
6. **Ecosystem** (Hadoop, Hive & Pig Integration)
7. **Unsupervised Scoring** (MiniBatchKMeans)
8. **Synthesis** (Engineering Results & Summary)

Problem Statement

Trigger-level Setting

- LHC bunch crossings @ **40 MHz** (25 ns interval)
- Online L1 trigger discards bulk before storage
- Reconstructed physics objects: $e^{\pm}$, $\mu^{\pm}$, jets, $E_{\mathrm{T}}^{\mathrm{miss}}$

Research Question

Can an auditable bounded-memory pipeline validate full HDF5 tensors, build typed Parquet tables, and rank atypical events on 6 CPU cores?

Pipeline Deliverables

- Quality gate: full-corpus zero-defect audit
- Spark SQL + RDD analytics on single-node Docker
- Unsupervised mini-batch distance scoring

646.4 MB HDF5 source
13.45M events verified

Dataset & Provenance

Property	Value
Repository	Zenodo 5046428 (DOI 10.5281/zenodo.5046428)
File	`background_for_training.h5` (646.39 MB)
MD5	`08cb4801229f9d480567f8d3f5cac134` ✓
SHA-256	`5f460bd3af44…cfe5`
Keys	`/Particles`, `_Names`, `_Classes`
Storage	float64, gzip-4, chunks (26274,1,1)

Raw Scale

$13{,}451{,}915 \times 19 \times 4 = 1{,}022{,}345{,}540$ scalars
$\times\, 8\,\text{B} = 8.18\,\text{GB}$ uncompressed ($\approx 12.7\times$ gzip ratio)

Audited Cardinality

- Dataset paper states **4,000,000** events
- File scan yields **13,451,915** rows
- Pipeline evaluates 100% without subsampling

SIGNAL BENCHMARKS (4 signals, 1 black-box)
Evaluated separately in literature

Fixed-Width Tensor Schema

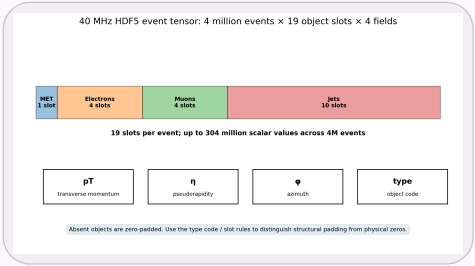


Figure 1: Documented 19×4 slot layout ([p_{T}, η, ϕ, type]).

Slots	Segment	Code
0	MET ($E_{\mathrm{T}}^{\mathrm{miss}}$)	$t{=}1$
1–4	electrons (max 4)	$t{=}2$
5–8	muons (max 4)	$t{=}3$
9–18	jets (max 10)	$t{=}4$
unused	zero-padding	$t{=}0$

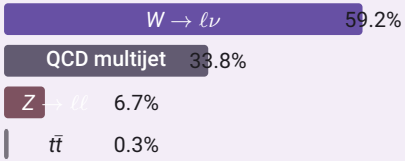
Validation Rules

- Objects sorted by descending p_{T} within group
- Padding rule: $t_i = 0 \Rightarrow p_{\mathrm{T}}{=}\eta{=}\phi{=}0.0$
- Type code gate: finite $\wedge$ integer $\in \{0, \ldots, 4\}$
- Schema assertion: `ndim==3`, shape[1:] $= (19, 4)$

Physics Context

Dataset & Schema

Standard Model process mixture (published)



Simulation & Selection

13 TeV *pp*, PYTHIA + DELPHES sim; single-lepton trigger selection.

Kinematics Conventions

$$p_{\mathrm{T}} = \sqrt{p_x^2 + p_y^2} \quad [\mathrm{GeV}], \quad \eta = -\ln[\tan(\theta/2)]$$

$$H_T = \sum_{\text{jets}} p_{\mathrm{T},i}, \quad S_T^{\ell} = \sum_{e,\mu} p_{\mathrm{T},i}$$

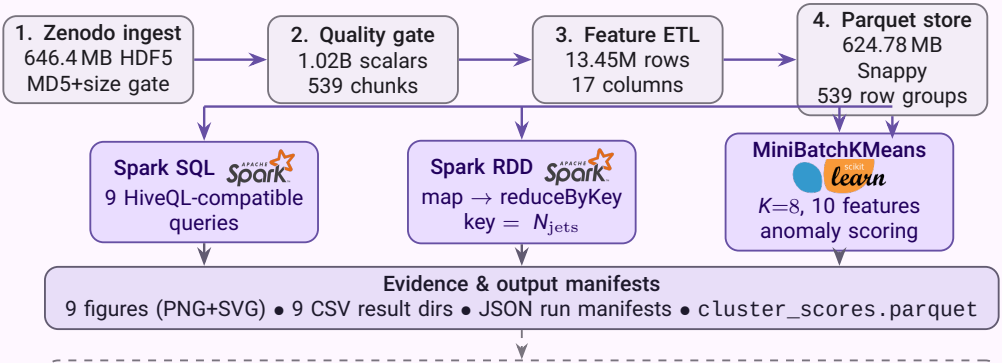
$E_{\mathrm{T}}^{\mathrm{miss}}$ magnitude read directly from fixed slot 0.

Species Mapping

$E_{\mathrm{T}}^{\mathrm{miss}}$ (1) · $e^{\pm}$ (2) · $\mu^{\pm}$ (3) · jet (4) · pad (0)

System Architecture

Pipeline



Row 1 = core path · Row 2 = parallel engines via distribution bus · Row 4 = external-infra artefacts

Methodology – Pipeline Architecture

Pipeline

Quality-first Design

1. Full checksum & provenance check
2. Schema assertion prior to ETL
3. 1:1 row correspondence throughout

Bounded-memory Execution

Fixed streaming chunks of 25,000 rows (539 row groups); peak container RSS ≤ 8 GB.

Parity Validation

- ETL: 13,451,915 in $=$ 13,451,915 out
- Spark: partition sum $=$ 13,451,915
- RDD: $\sum n_j = 13{,}451{,}915$
- ML: 100% rows scored without loss

Timings: Spark 93.8 s · ML 25.2 s (single host, 6 cores).

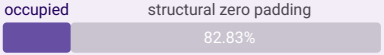
Data Quality Gate (Full-Corpus Audit)

Quality & Distributions

Defect Counters (1.02B Scalars Scanned)

Integrity Check	Count
NaN entries	0
$+\infty$ / $-\infty$ entries	0 / 0
Invalid type code $\notin \{0..4\}$	0
Non-integral type code	0
Negative occupied p_{T}	0
$\lvert\phi\rvert > \pi$ out-of-range	0
Nonzero padding kinematics	0
Misplaced slot code	0
Total defects	**0**

Slot Occupancy (255.6M slots)

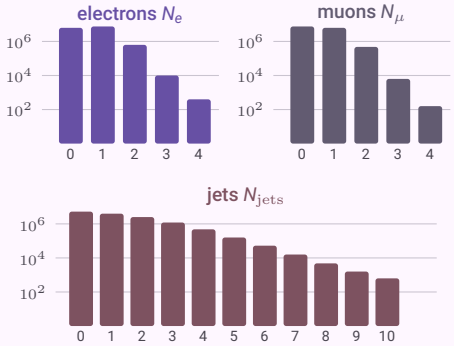


100% of events contain valid zero-padding.

0 defects
100% clean data gate

Object Multiplicity (All 13.45M Events)

Quality & Distributions



Log-scale multiplicity. $N_{\mathrm{jets}} \leq 2$ covers **86.0%** of events; tail reaches $N_{\mathrm{jets}}{=}10$ (731 events).

Transverse-Momentum Spectra by Object Class

Quality & Distributions

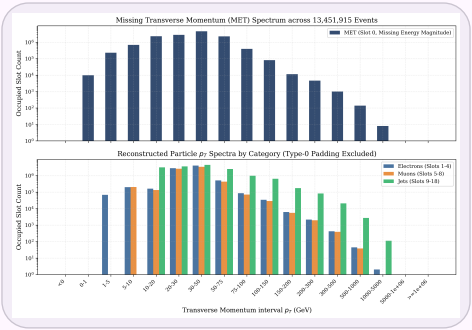


Figure 2: Log-scale p_{T} histograms per particle class (all 43.88M occupied slots). Trigger threshold $p_{\mathrm{T}} > 23$ GeV visible; spectra peak at 20–50 GeV.

MET vs Jet Activity (Joint Density)

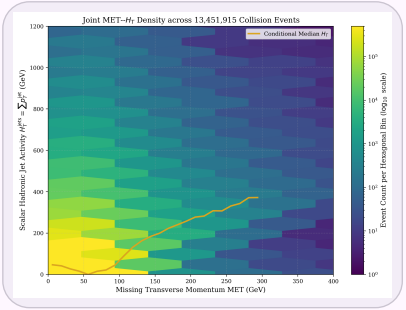


Figure 3: Log-scale 2D hexbin density across all 13.45M events.

Phase Space Characteristics

- Core bulk: $E_{\mathrm{T}}^{\mathrm{miss}} < 50\,\mathrm{GeV} \wedge H_T < 100\,\mathrm{GeV}$
- Conditional median H_T increases with $E_{\mathrm{T}}^{\mathrm{miss}}$
- Sparse high-activity tail flagged for review

2D BIN MATRIX (Q04 output)
943 populated density bins

Scalar Jet Activity Conditioned on N_{jets}

Quality & Distributions

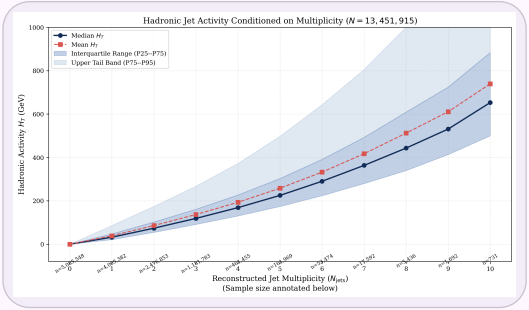


Figure 4: H_T distribution conditioned on jet multiplicity (0–10).

Multiplicity Trends

- Median H_T scales from 0 to 652.9 GeV
- Spread expands at higher jet counts
- Exactly zero H_T for 5.08M zero-jet events

652.9 GeV median H_T at $N_{\text{jets}}=10$

Feature Correlations

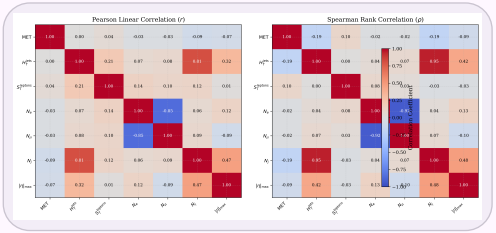


Figure 5: Feature correlations (13.45M rows).

Key Coefficients

- H_T–N_{jets}: r=0.74, ρ=0.87
- $E_{\text{T}}^{\text{miss}}$–$H_T$: r=0.15 (orthogonal)

Interpretation

H_T–N_{jets} correlation is construction-driven ($H_T = \sum p_{\text{T}}$). Weak $E_{\text{T}}^{\text{miss}}$–$H_T$ correlation confirms independent observables.

Full-Corpus H_T Distribution

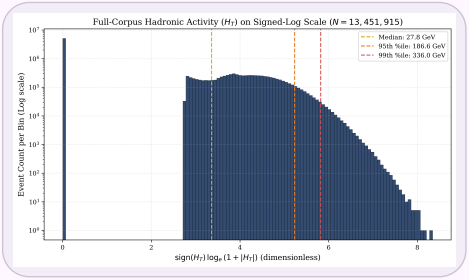


Figure 6: H_T distribution with signed-log axis.

Distribution Characteristics

- Exact delta spike: 5.08M events at $H_T = 0$
- Physical tail extends beyond 4,000 GeV
- Log-scale display captures dynamic range

Spark SQL Query Suite

Spark Analytics

ID	Script	Key Operations	Rows
Q01	`01_dataset_overview.sql`	COUNT, SUM, AVG, `percentile_approx`	1
Q02	`02_object_multiplicity.sql`	Multiplicity distributions (e, μ, jets)	21
Q03	`03_ht_by_jet_multiplicity.sql`	Conditional quartiles & Pearson correlation	11
Q04	`04_met_ht_density_bins.sql`	2D binning (20 GeV $\times$ 50 GeV cells)	943
Q05	`05_quality_flags.sql`	Quality gate flag verification	1
Q06	`06_high_activity_candidates.sql`	Extreme activity event candidates (H_T top 100)	100
Q07	`07_feature_correlation.sql`	Grouped feature correlations vs N_{jets}	11
Q08	`08_window_jet_slice_rankings.sql`	Window functions (`DENSE_RANK`) top-3 per slice	30
Q09	`09_multidimensional_rollup.sql`	Multidimensional grouping (`ROLLUP`)	28

Runtime Execution Contract

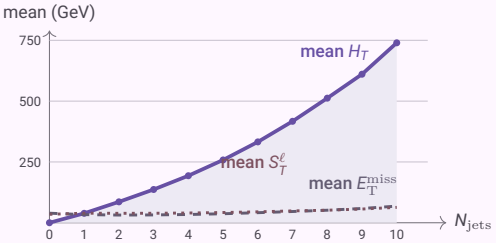
Spark 3.5.7 `local[6]` · 12 partitions · AQE on · **93.8 s** suite runtime.

✓ Parity Verification

All queries read typed Parquet with 1:1 row parity against raw audit.

MapReduce Key-Value Aggregation (RDD)

Spark Analytics



RDD `map` → `reduceByKey` means (13.45M events).

Algorithm Specification

$$\mathrm{Map}(e) \mapsto \left(N_{\mathrm{jets}},\ (1, H_T, E_{\mathrm{T}}^{\mathrm{miss}}, S_T^\ell)\right)$$

$$\mathrm{Reduce}: (n, s_H, s_M, s_L) \oplus (n', s'_H, s'_M, s'_L) = (n{+}n',\ s_H{+}s'_H,\ s_M{+}s'_M,\ s_L{+}s'_L)$$

Validation

$\sum_j n_j = 13{,}451{,}915 = N_{\mathrm{input}}$ ✓

Portability

Same protocol in `hadoop/mapper.py` and `reducer.py` for HDFS/YARN.

Ecosystem Artefacts (Hadoop · Hive · Pig)

Ecosystem



Hadoop Streaming

`hadoop/mapper.py` emits `n_jets → (1, HT, MET, S_l)`
`hadoop/reducer.py` sums per key, emits means.

HADOOP STREAMING
MapReduce reference



Hive DDL

```
hive/register_events.hql:
CREATE DATABASE ca3_40mhz;
CREATE EXTERNAL TABLE ...
PARTITIONED BY (n_jets);
STORED AS PARQUET;
```

HIVESERVER2 RUN
External metastore



Pig Latin 0.17

```
pig/analyze_events.pig:
LOAD 5-col TSV →
GROUP BY n_jets →
FOREACH ... COUNT, AVG →
ORDER BY → STORE
```

PIG LOCAL RUN
Local profile runner

Spark SQL executes identical queries in core stack; scripts provide standalone portability.

Streaming MiniBatchKMeans

Unsupervised Scoring

Model Architecture (10 Features)

$E_{\mathrm{T}}^{\mathrm{miss}}$, H_T, S_T^{ℓ}, N_e, N_μ, N_{jets}, lead e/μ/jet p_{T}, $\max|\eta|$

1. `StandardScaler.partial_fit` → corpus μ_j, σ_j
2. `MiniBatchKMeans.partial_fit`, $K{=}8$, rs$=27$
3. Predict + distance: $s(\mathbf{x}) = \min_k \|\mathbf{z}(\mathbf{x}) - \boldsymbol{\mu}_k\|_2$

Full-Corpus Coverage

Rows scaled = fit = scored = 13,451,915 (100%); 25.2 s runtime.

Cluster Population ($K{=}8$)



Largest: cluster 1 (3.99M) · Smallest: cluster 4 (282k).

Bounded Memory

`partial_fit` streams 25k chunks without loading full corpus into RAM.

Distance Outlier Profiles

Unsupervised Scoring

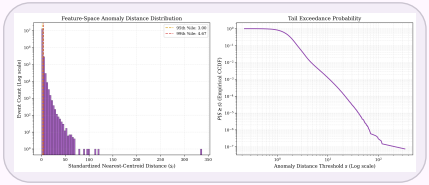


Figure 7: Distance distribution (full corpus).

Min 0.224 · Mean 1.648 · $\sigma = 0.95$
Max 336.8 · P99 = 4.62

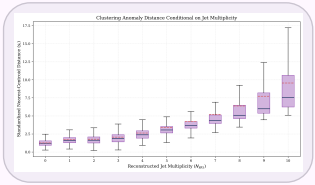


Figure 8: Distance conditioned on N_{jets}.

Geometric Insight

Median distance grows with N_{jets} ($1.15 \rightarrow 7.12$). Scores isolate complex multi-jet events for targeted expert inspection.

Evidence Classes & Quality Guarantees

Synthesis

Four Evidence Classes

- **P** **Published Fact**: Zenodo paper / record
- **C** **Calculated Quantity**: Exact verified arithmetic
- **L** **Local Measurement**: JSON manifest values
- **A** **Conditional Artifact**: Generator script outputs

Implemented Guarantees

- Download gates on MD5 and exact byte count
- Schema assertion precedes ETL transform
- 1:1 row correspondence verified across engines
- Unit test assertions on synthetic fixtures

CA3 Course-Outcome Alignment

Synthesis

Course Concept	Evidence in Project	Scope Note
MapReduce	Spark RDD key-value map/reduceByKey, count parity, Hadoop Streaming scripts	Native Hadoop requires external cluster
Hive / SQL	9 HiveQL queries, partitioned Parquet, grouped stats, Window + ROLLUP	Spark SQL is primary runtime
Pig Latin	GROUP, FOREACH, COUNT, AVG local mode over TSV	Pig 0.17 profile
Spark Columnar	HDF5→Parquet ETL, partitionBy, read-back parity	Single-node Docker
Intelligent Analysis	MiniBatchKMeans novelty score over 10 standardized features	scikit-learn streaming
Visual Analytics	9 figures: spectra, density, conditionals, correlations, distances	Full-corpus basis

Conclusion & Caveats

Synthesis

Verified Outcomes

- 13,451,915 events audited $\rightarrow$ 0 defects
- ETL parity $\rightarrow$ 624.78 MB Parquet, 539 row groups
- 9 SQL groups + RDD aggregation $\sum n_j = N$ ✓
- $K{=}8$ full-corpus distance scores
- 9 figures backed by machine manifests

Operational Notes

Single-node `local[6]` executes all queries in 93.8 s. Upper tail ranks multi-jet topologies for expert inspection.

Key Numbers

1,022,345,540 scalars scanned
82.83% structural padding
86.0% events with $N_{\text{jets}} \leq 2$
652.9 GeV median H_T @ $N_{\text{jets}}{=}10$
4.62 P99 centroid distance
93.8 s Spark suite runtime

FUTURE WORK
Signal benchmarks · Multi-node scaling

References & Reproducibility

Synthesis

References

- E. Govorkova et al., LHC dataset, *Sci. Data* 9:118, 2022.
- CERN / Fermilab, Training Dataset, Zenodo 5046428, 2021.
- ML for Particle Physics, ADC 2021 Benchmark, 2021.
- Apache Software Foundation, Spark 3.5.7 Guides, 2024.
- Apache Hadoop · Hive DDL · Pig 0.17 Docs.
- Apache Arrow / PyArrow Docs (v18.1.0), 2024.
- F. Pedregosa et al., Scikit-learn, *JMLR* 12:2825, 2011.

Reproduce Entire Pipeline

```
cp .env.example .env
make build && make test && make all
make check-quality
```

Key Verified Results

- 13.45M events verified with zero scalar defects
- Exact row match across Parquet, Spark & RDD
- 93.8 s single-node Docker reproducibility
- Full 100% unsupervised streaming outlier ranks